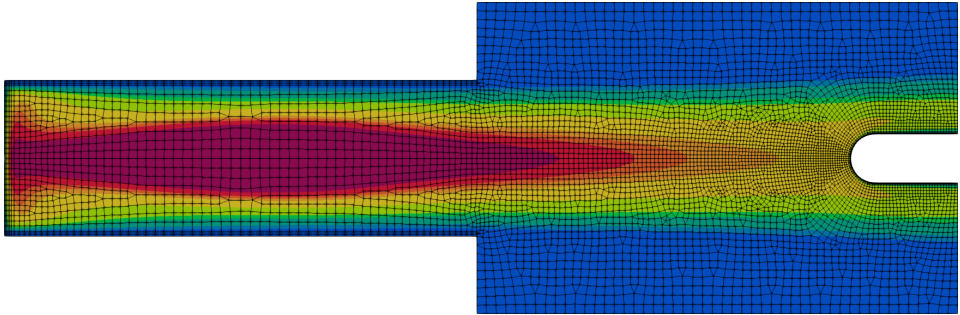


A Hybridized Discontinuous Galerkin Solver for Inductively Coupled Plasma

Nicolas Corthouts

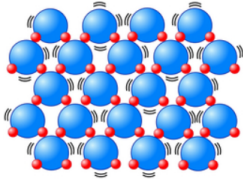


Introduction

Etats de la matière: eau à pression atmosphérique

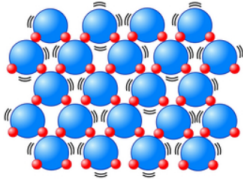
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Solide

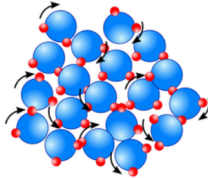


Etats de la matière: eau à pression atmosphérique

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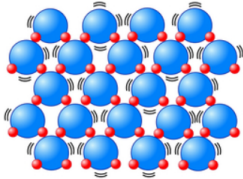


Liquide

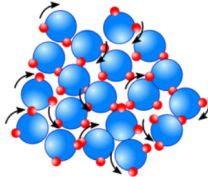


Etats de la matière: eau à pression atmosphérique

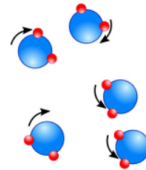
Solide



Liquide

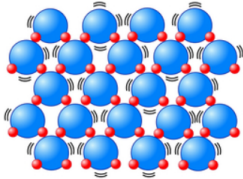


Gaz

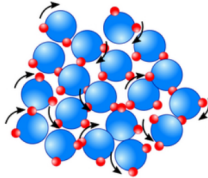


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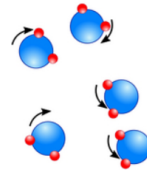
Solide



Liquide

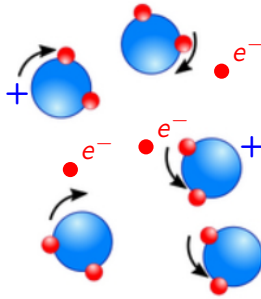


Gaz



Que se passe-t-il si on chauffe encore plus?

Plasma: le quatrième état de la matière

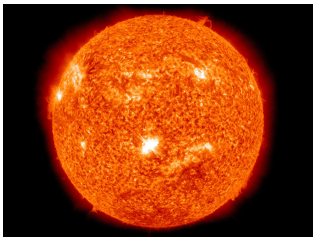


Un plasma est un gaz quasi-neutre composé de particules chargées et neutres démontrant un comportement collectif.¹

¹F. F. Chen. Introduction to Plasma Physics and Controlled Fusion. Ed. by Springer International Publisher. 2016

Plasma dans la vie de tous les jours

Les plasmas composent 90% de l'univers visible.



Fusion nucléaire, médecine, métallurgie, lasers, création de microprocesseurs, ...

²Courtesy of Wikipedia.

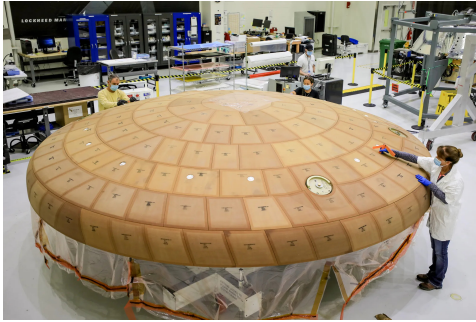
Plasma en réentrée atmosphérique



Vitesse de réentrée $\simeq 10 \text{ km s}^{-1}$: choc transformant l'air en plasma.

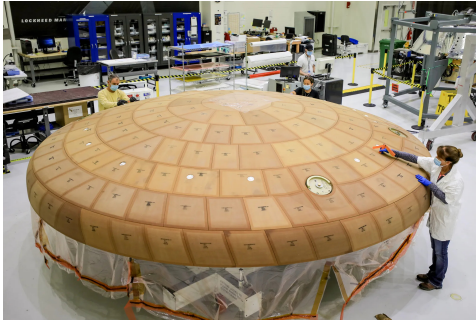
³Coutesy of NASA.

Système de protection thermique et destruction des déchets spatiaux



TPS

Système de protection thermique et destruction des déchets spatiaux

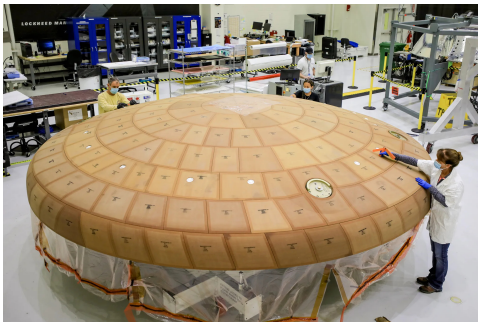


TPS



Déchets spatiaux

Système de protection thermique et destruction des déchets spatiaux



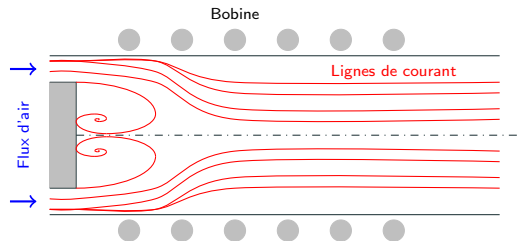
TPS



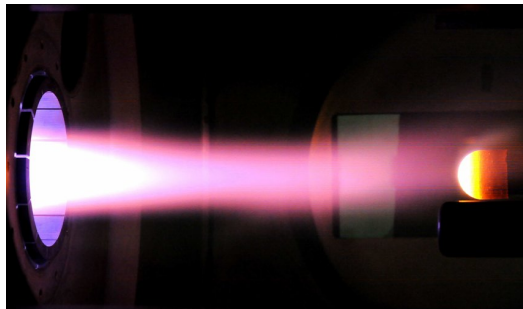
Déchets spatiaux

Développement de dispositifs expérimentaux reproduisant les plasmas de réentrée atmosphérique.

Plasma à induction



Torche

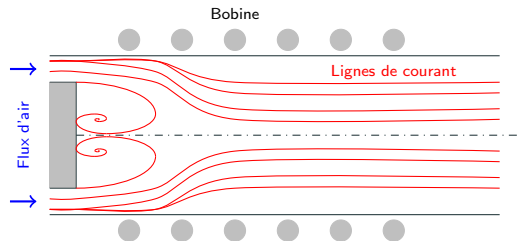


Chambre

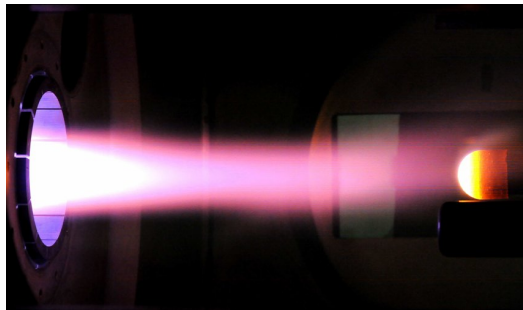
Exemples: Plasmatron (VKI), Plasmatron X (Illinois), IPG (Russie), ...

Basé sur le principe de **transfer de chaleur local**.

Plasma à induction



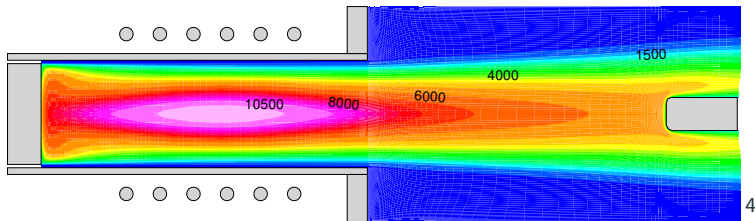
Torche



Chambre

Résoudre les équations de Maxwell, de Navier-Stokes + modèles physico-chimiques + modèle de radiation.

Des solveurs numériques (volumes finis) ont été développés afin de préparer au mieux les expériences dans les torches à induction.

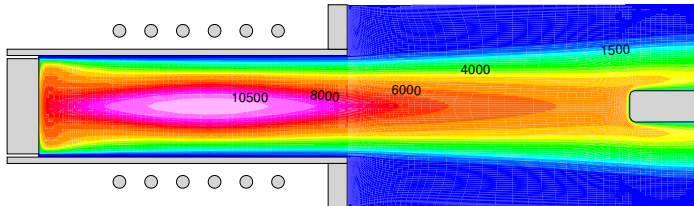


⁴Courtesy of Thierry Magin.

La plupart des solvers actuels

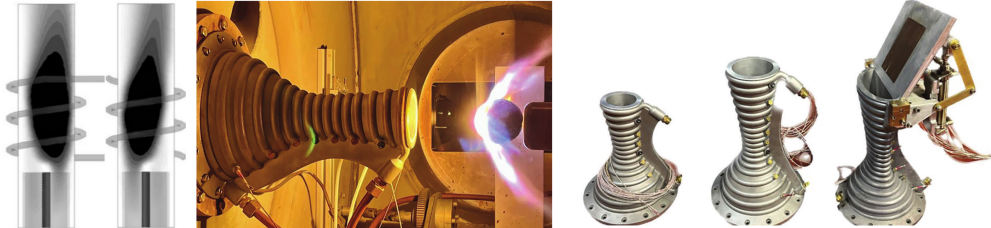
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- Ne représentent que des géométries axisymétriques et simples.



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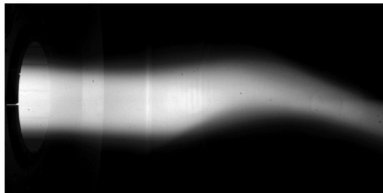
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- Utilisent un modèle à l'équilibre chimique et thermique sans radiation.

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- Sont en régime établi.

Besoin d'un solveur instationnaire.

- Utilisent un modèle à l'équilibre chimique et thermique sans radiation.

C'est faux! Car les déséquilibres chimique et thermique ont été observés dans les plasmas à induction! La radiation joue aussi un rôle important dans le transfert de chaleur.

La plupart des solveurs actuels

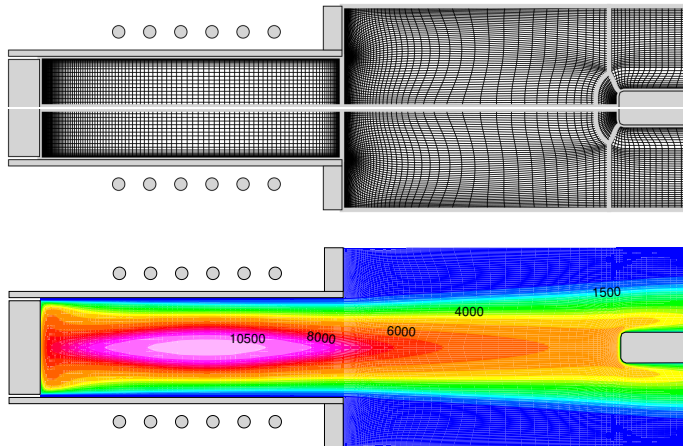
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Besoin d'un solveur avec une chimie plus détaillée et de la radiation.

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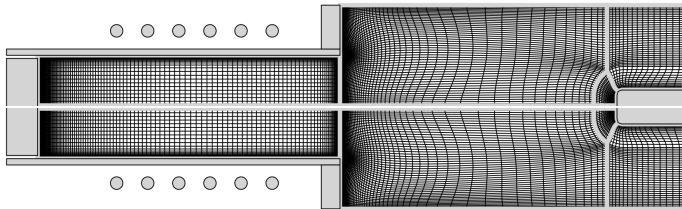
Peut-on améliorer les solveurs volume finis actuels afin de pouvoir prendre en compte toute la physique?

Le problème des solvers volume finis actuels.



Pour les volumes finis, la solution est constante sur chaque éléments.

Le problème des solvers volume finis actuels.

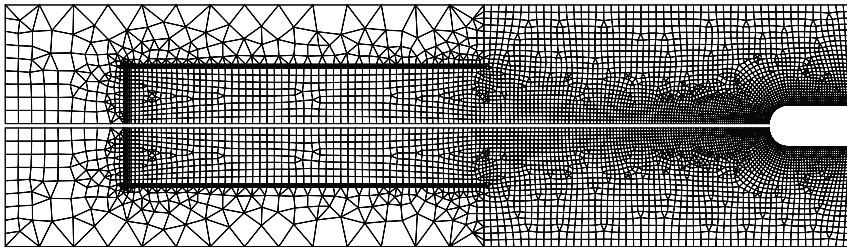


Les volumes finis nécessitent un maillage très fin dans les régions de grande variation ou de géométrie complexe.

- Demande beaucoup d'attention au maillage.
- Quasi impossible pour les géométries complexes.
- Demande un maillage trop fin partout pour une physique plus complexe.

Développer un solveur faisant partie des méthodes de Galerkin discontinues pour les plasma à induction.

- La solution n'est plus constante sur chaque élément.
- Beaucoup plus grande flexibilité de maillage.



Research questions

With the objective of producing a new high order solver for inductively coupled plasma, we will try to answer the following questions:

- Q1 In addition of being precise, can a high-order solver be robust for inductively coupled plasma applications?**
- Q2 Is the developed solver user-friendly?**
- Q3 Can the new solver be easily adapted to the new experiments performed in ICP facilities?**

Model for inductively coupled plasma

A plasma is a quasi-neutral gas composed of charged and neutral particles exhibiting a collective behaviour.

Collision = particles interaction

Short ranged



Direct contact (local & binary)

Collision = particles interaction

Short ranged



Direct contact (local & binary)

Long ranged



Electric force (distance and collective)

Plasma collective behaviour

Collision = particles interaction

Short ranged



Direct contact (local & binary)

Long ranged



Electric force (distance and collective)

On sufficiently large scales ($> 1 \mu\text{m}$ in our case), the electric neutrality of the plasma is guaranteed by the long range collisions (electric force).

Collisions may lead to chemical reactions.

If τ_{chem} and τ_{hydro} are characteristic chemical reaction and flow time respectively:

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$$\tau_{chem} \gg \tau_{hydro}$$

Chemical equilibrium

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$\tau_{chem} \gg \tau_{hydro}$	$\tau_{chem} \simeq \tau_{hydro}$
Chemical equilibrium	Chemical non-equilibrium

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Plasmas are in chemical equilibrium in this work.

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Thermal non-equilibrium

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Inneficient collisions

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Thermal non-equilibrium

Efficient collisions

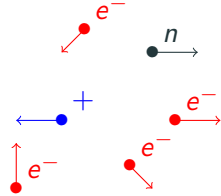
$$T_e \simeq T_h$$

Thermal equilibrium

Plasmas are in thermal equilibrium in this work.

Hypothesis on the plasma

- Collision-dominated and thermal plasma



$$T_e \simeq T_h$$

Hypothesis on the plasma

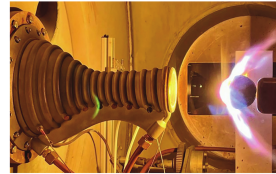
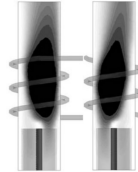
- Collision-dominated and thermal plasma
- Quasi-neutrality

Hypothesis on the plasma

- Collision-dominated and thermal plasma
- Quasi-neutrality
- No displacement current
- Electrons return fast to equilibrium when perturbed by ELM waves.
- ELM wavelength $\gg$ plasma length scale.

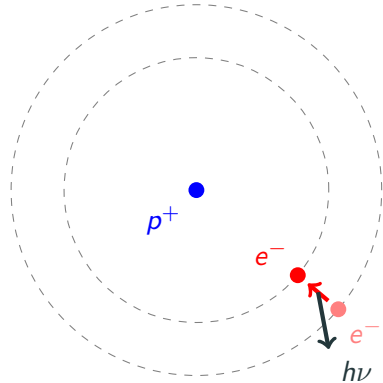
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- Optically thick plasma
- Local thermodynamic equilibrium

Thermal + chemical equilibrium =
LTE

Hypothesis on the plasma

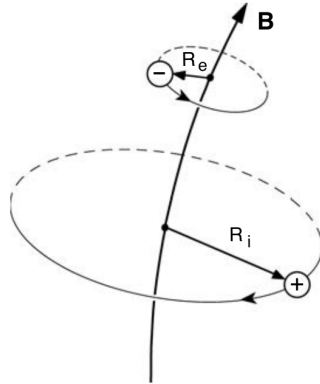
- Collision-dominated and thermal plasma
- Quasi-neutrality
- No displacement current
- Axisymmetry
- Optically thick plasma
- Local thermodynamic equilibrium
- No elemental de-mixing

Elemental de-mixing = diffusion of elements for LTE.

Results closer to non-equilibrium.

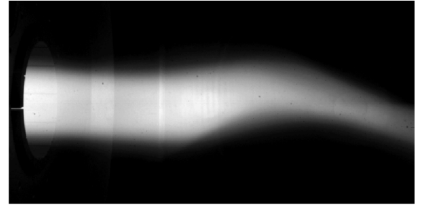
Hypothesis on the plasma

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- Steady-state



Equations averaged over one period of the induction current.

Hypothesis on the plasma

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- Axisymmetry
- Optically thick plasma
- Local thermodynamic equilibrium
- No elemental de-mixing
- Unmagnetized plasma
- Steady-state

The electric field:

- It is ambipolar ($j_z = j_r = 0$).
- The coils are thin parallel wires surrounding the facility.
- Axisymmetric
- $E_{tot} = E_C + E_P$
- We use phasor notation.

Equations governing the plasma

Navier-Stokes equations

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v} \mathbf{v}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \mathbf{F}_L$$

$$\partial_t \left(\rho e + \frac{1}{2} \rho \|\mathbf{v}\|^2 \right) + \nabla \cdot \left(\rho e \mathbf{v} + \frac{1}{2} \rho \|\mathbf{v}\|^2 \mathbf{v} + p \mathbf{v} \right) = \nabla \cdot (\boldsymbol{\tau} \mathbf{v}) + \nabla \cdot \mathbf{q} + P_J$$

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$$\boldsymbol{\tau} = \eta \left(\nabla \mathbf{v} + \nabla \mathbf{v}^T \right) - \frac{2}{3} \eta \nabla \cdot \mathbf{v}$$

$$\mathbf{q} = -k \nabla T$$

$$F_z^L = \frac{\sigma_e}{4\pi f} \left[E_l^{Im} \partial_z E_l^{Re} - E_l^{Re} \partial_z E_l^{Im} \right]$$

$$F_r^L = \frac{\sigma_e}{4\pi f} \left[E_l^{Im} \frac{1}{r} \partial_r (r E_l^{Re}) - E_l^{Re} \frac{1}{r} \partial_r (r E_l^{Im}) \right]$$

$$P^J = \frac{\sigma_e}{2} \left[(E_l^{Im})^2 + (E_l^{Re})^2 \right]$$

Electric field equation

$$\partial_{zz}^2 E_P + \frac{1}{r} \partial_r (r \partial_r E_P) - \frac{E_P}{r^2} = i 2 \pi f \mu_0 \sigma_e (E_C + E_P)$$

Electric field equation

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μ_0 : magnetic permeability

σ_e : electron electric conductivity

f : induction frequency

Electric field equation

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$$E_C = \sum_{l=1}^{N_{coil}} i f \mu_0 I_C \sqrt{\frac{r_0}{r}} \left[2 \frac{E_2(k_l)}{k_l} - E_1(k_l) \left(\frac{2}{k_l} - k_l \right) \right]$$

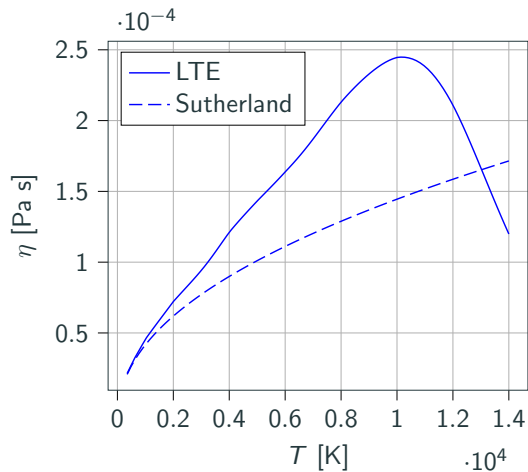
$E_1(k)$, $E_2(k)$ elliptic integral of the first and second kind.

M⁺⁺utation

Multicomponent Thermodynamic And Transport properties for IONized gases in C++

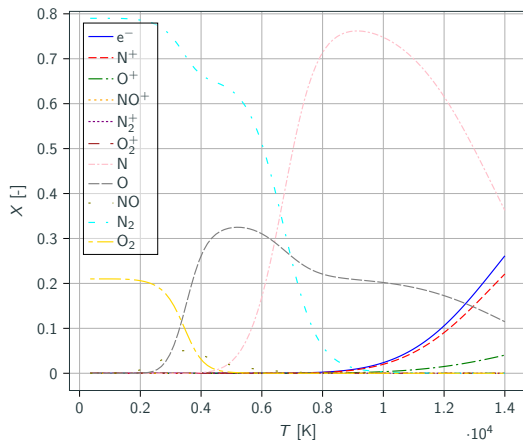
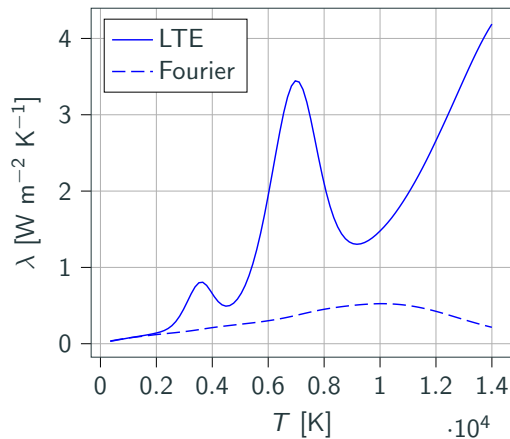
Uses a Galerkin approximate solution of the linearized Boltzmann equation for computing the transport coefficients $(\sigma_e, \eta, \lambda)$.

Transport properties : viscosity

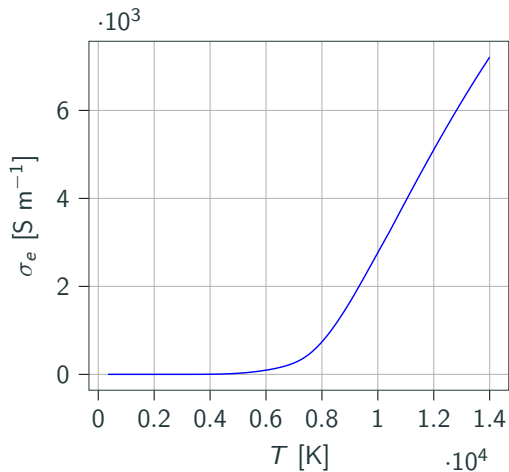


$$\eta_i \propto \frac{\sqrt{T_i}}{\sigma_i}.$$

Transport properties : heat conductivity



Transport properties : electric conductivity



Navier-Stokes

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Maxwell

$$\partial_{zz}^2 E_P + \frac{1}{r} \partial_r (r \partial_r E_P) - \frac{E_P}{r^2} = i 2 \pi f \mu_0 \sigma_e (E_C + E_P)$$

Coupling of the systems

Navier-Stokes

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$$

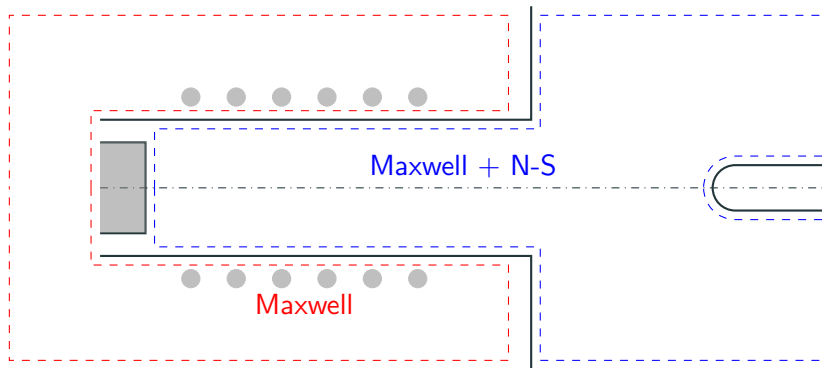
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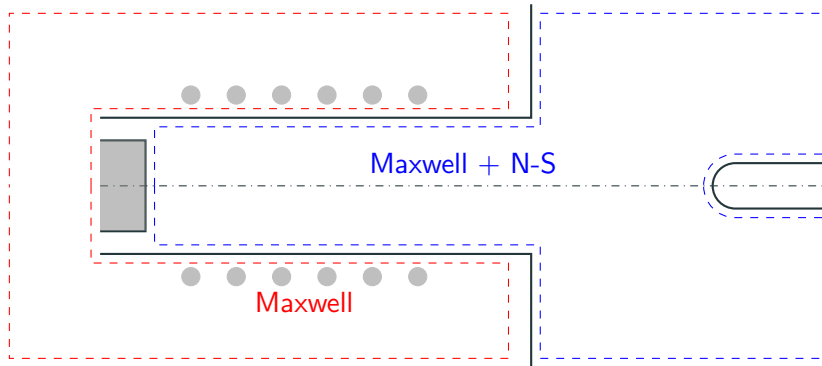
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Computational domain

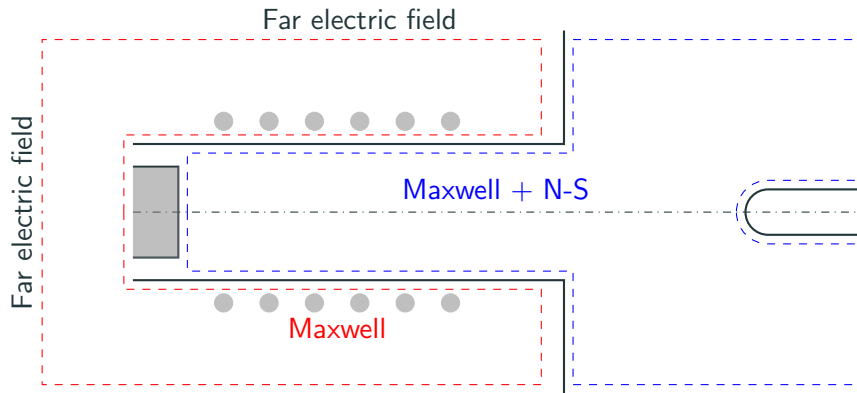


Computational domain



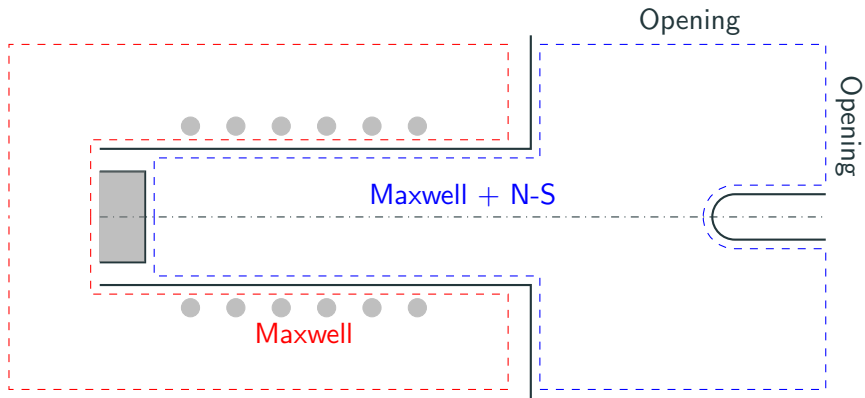
Symmetry axis $\partial_r p = 0$, $\partial_r u_z = 0$, $u_r = 0$, $\partial_r T = 0$ and $E_P = 0$.

Computational domain



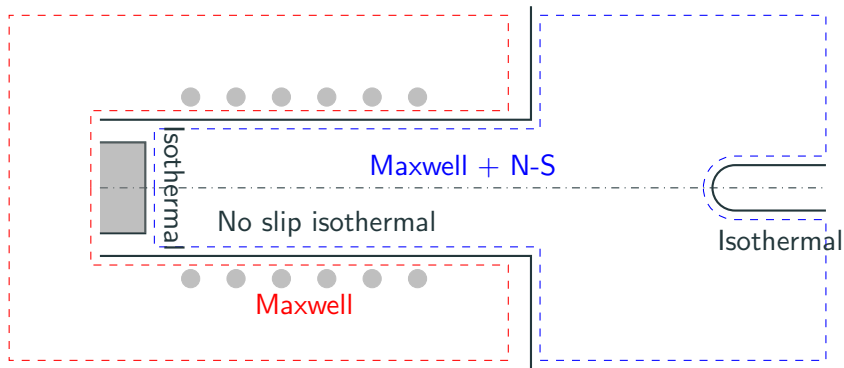
Vanishing far electric field $E_P = 0$.

Computational domain



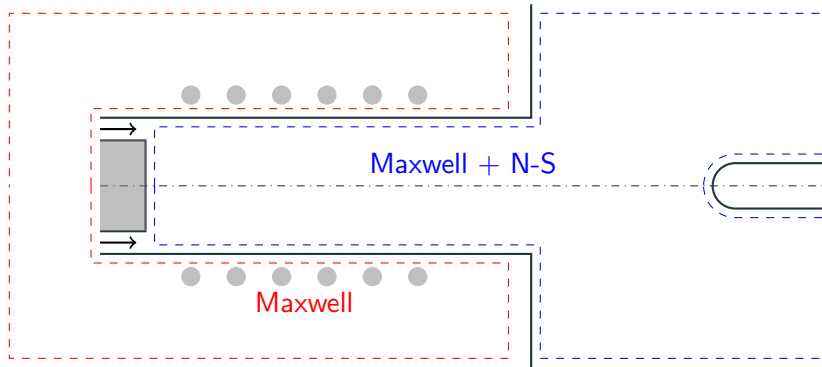
Openings $p = p_0$

Computational domain



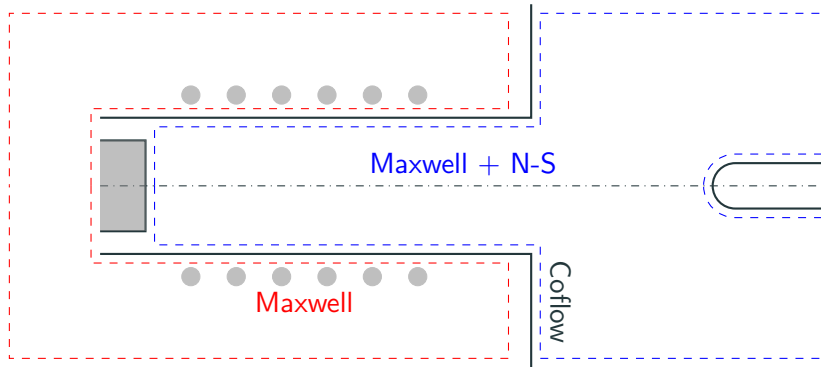
No-slip wall isothermal $T = T_{wall}$ and $\mathbf{u} = \mathbf{0}$.

Computational domain

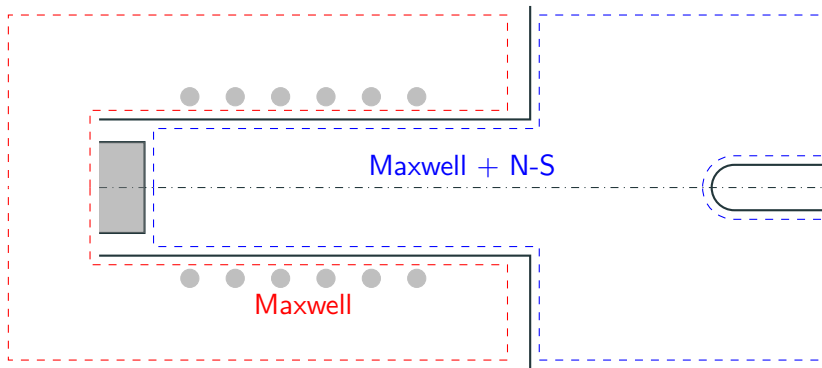


Inflow $u_z = U_{in}$, $u_r = 0$ and $T = T_{inlet}$.

Computational domain



Stabilizing co-flow $T = T_{wall}$, $\mathbf{u} = u_{coflow}\mathbf{e}_z$

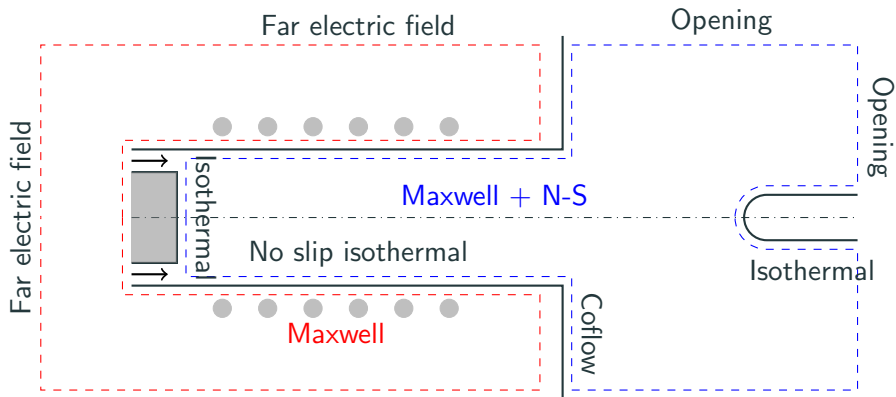


ICP and Maxwell interface

$$\nabla E_p^{Re,NS} \cdot \mathbf{n}^{NS} = \nabla E_p^{Re,MAX} \cdot \mathbf{n}^{MAX}$$

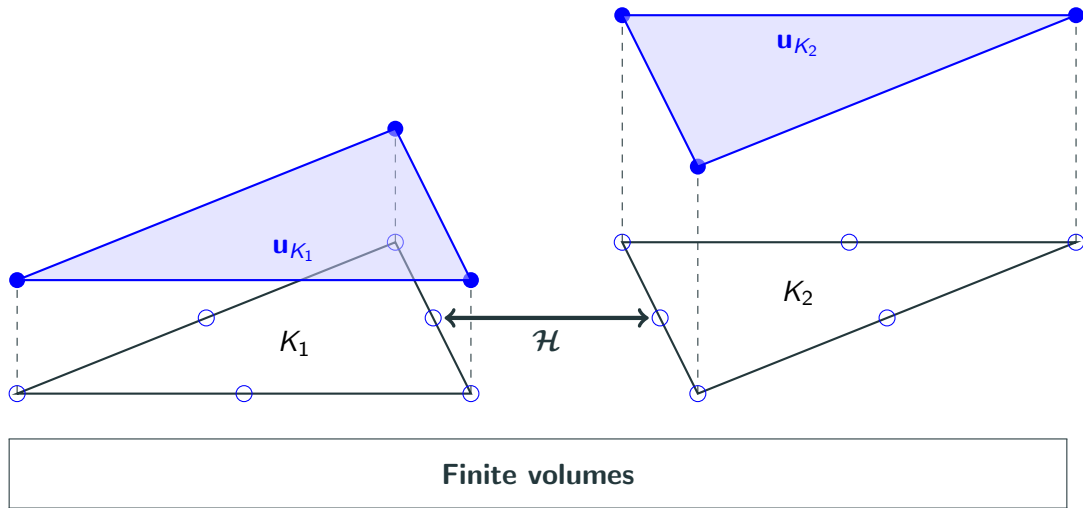
$$\nabla E_p^{Im,NS} \cdot \mathbf{n}^{NS} = \nabla E_p^{Im,MAX} \cdot \mathbf{n}^{MAX}$$

Computational domain

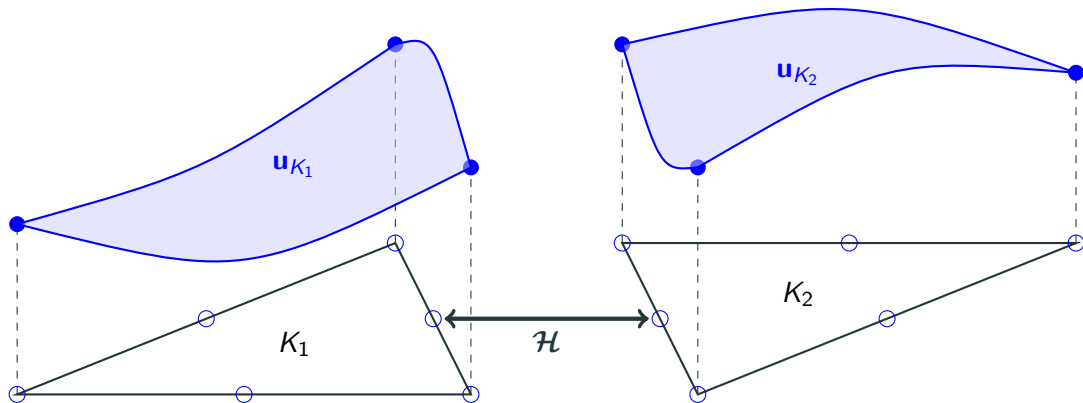


Hybridized discontinuous Galerkin method

Finite volume and (hybridized) discontinuous Galerkin

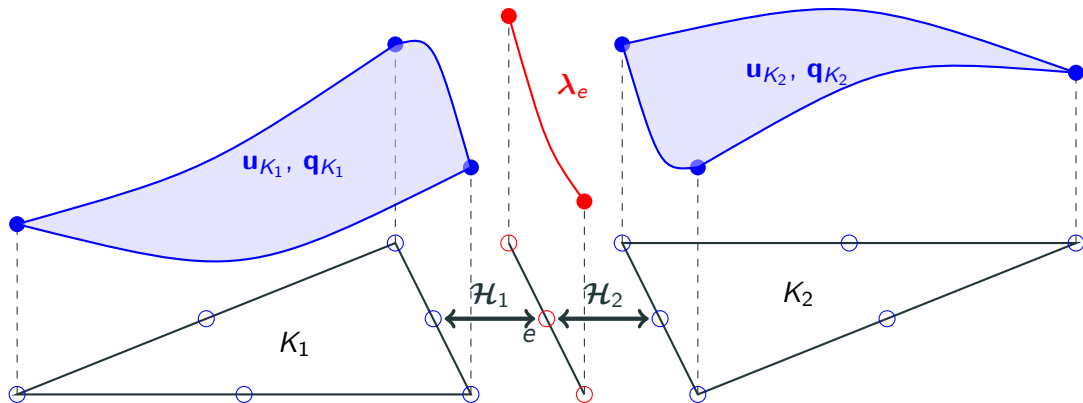


Finite volume and (hybridized) discontinuous Galerkin



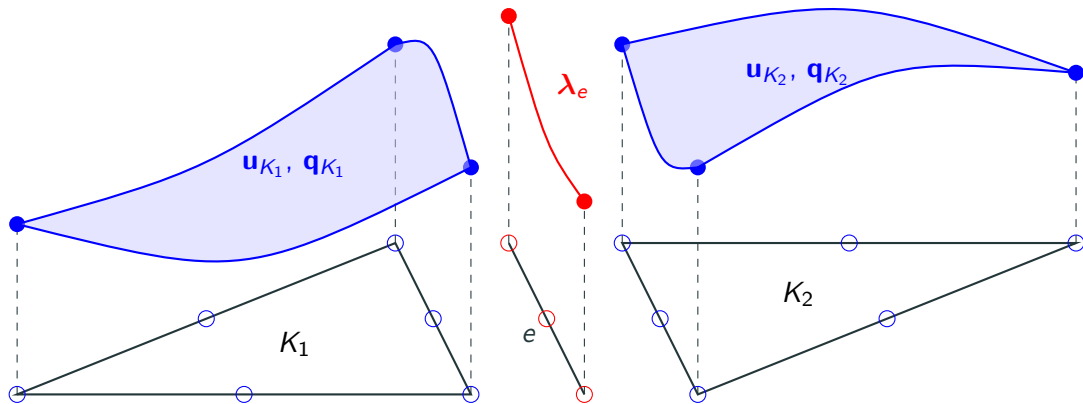
Discontinuous Galerkin

Finite volume and (hybridized) discontinuous Galerkin



Hybridized discontinuous Galerkin

Finite volume and (hybridized) discontinuous Galerkin



Why HDG over DG? As will be seen later, HDG allows for static condensation, effectively reducing the number of DOFs when using a Newton solver.

Conservative form of the equations

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v} \mathbf{v} + p \mathbb{I}) - \nabla \cdot \boldsymbol{\tau} = \mathbf{F}_L$$

$$\partial_t (\rho E) + \nabla \cdot (\rho E \mathbf{v} + p \mathbf{v}) - \nabla \cdot (\boldsymbol{\tau} \mathbf{v} + \mathbf{q}) = P_J$$

$$- \nabla \cdot (\nabla E_P) = -\frac{E_P}{r^2} - i2\pi f \mu_0 \sigma_e (E_C + E_P)$$

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$$- \nabla \cdot (\nabla E_P) = -\frac{E_P}{r^2} - i2\pi f \mu_0 \sigma_e (E_C + E_P)$$

$$\partial_t \mathbf{w}(\mathbf{u}) + \nabla \cdot \mathbf{F}_c(\mathbf{u}) - \nabla \cdot \mathbf{F}_v(\mathbf{u}, \nabla \mathbf{u}) = \mathbf{S}(\mathbf{u}, \nabla \mathbf{u})$$

Conservative form of the equations

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v} \mathbf{v} + p \mathbb{I}) - \nabla \cdot \boldsymbol{\tau} = \mathbf{F}_L$$

$$\begin{aligned} \partial_t (\rho E) + \nabla \cdot (\rho E \mathbf{v} + p \mathbf{v}) - \nabla \cdot (\boldsymbol{\tau} \mathbf{v} + \mathbf{q}) &= P_J \\ - \nabla \cdot (\nabla E_P) &= -\frac{E_P}{r^2} - i2\pi f \mu_0 \sigma_e (E_C + E_P) \end{aligned}$$

$$\partial_t \mathbf{w}(\mathbf{u}) + \nabla \cdot \mathbf{F}_c(\mathbf{u}) - \nabla \cdot \mathbf{F}_v(\mathbf{u}, \nabla \mathbf{u}) = \mathbf{S}(\mathbf{u}, \nabla \mathbf{u})$$

Model problem

$$\begin{aligned}\partial_t \mathbf{w}(\mathbf{u}) + \nabla \cdot (\mathbf{F}_c(\mathbf{u}) - \mathbf{F}_v(\mathbf{u}, \nabla \mathbf{u})) &= \mathbf{S}(\mathbf{u}, \nabla \mathbf{u}) && \text{on } \Omega \\ \mathbf{u} &= \mathbf{u}_{bc} && \text{on } \partial\Omega_D \\ \mathbf{F}_v \cdot \mathbf{n} &= \mathbf{F}_{v,n,bc} && \text{on } \partial\Omega_N \\ \mathbf{u}(t=0) &= \mathbf{U} && \text{on } \Omega\end{aligned}$$

$\mathbf{F}_{c,v}$: convective and diffusive fluxes.

$\mathbf{S}$: source terms.

$\mathbf{w}$: conservative variables.

$\mathbf{u}, \mathbf{u}_{bc}$: solution and boundary condition.

Ω : domain of the problem.

$\partial\Omega_{D,N}$: domain boundary for Dirichlet and Neumann BC.

$\mathbf{U}$: initial data.

Model problem

$$\begin{aligned}\partial_t \mathbf{w}(\mathbf{u}) + \nabla \cdot (\mathbf{F}_c(\mathbf{u}) - \mathbf{F}_v(\mathbf{u}, \nabla \mathbf{u})) &= \mathbf{S}(\mathbf{u}, \nabla \mathbf{u}) && \text{on } \Omega \\ \mathbf{u} &= \mathbf{u}_{bc} && \text{on } \partial\Omega_D \\ \mathbf{F}_v \cdot \mathbf{n} &= \mathbf{F}_{v,n,bc} && \text{on } \partial\Omega_N \\ \mathbf{u}(t=0) &= \mathbf{U} && \text{on } \Omega\end{aligned}$$

Before going further, some notation:

$$\begin{aligned}(a, b)_K &= \int_K ab \, dV, \\ \langle a, b \rangle_{\partial K} &= \int_{\partial K} ab \, dS.\end{aligned}$$

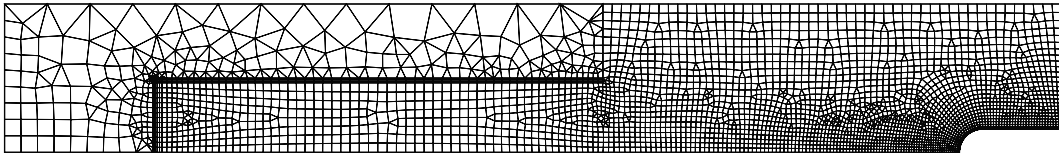
$$(\partial_t \mathbf{w} - \mathbf{S}, w)_\Omega - (\mathbf{F}_c - \mathbf{F}_v, \nabla w)_\Omega + \langle (\mathbf{F}_c - \mathbf{F}_v) \cdot \mathbf{n}, w \rangle_{\partial\Omega} = 0, \quad \forall w \in L_2(\Omega)$$

Multiplying by a function $w \in L_2(\Omega)$ and integrating over the domain gives the **weak form**.

From weak form to HDG

$$(\partial_t \mathbf{w} - \mathbf{S}, w)_{\mathcal{T}} - (\mathbf{F}_c - \mathbf{F}_v, \nabla w)_{\mathcal{T}} + \sum_{K \in \mathcal{T}} \langle (\mathbf{F}_c - \mathbf{F}_v) \cdot \mathbf{n}, w \rangle_{\partial K} = 0, \quad \forall w \in L_2(\Omega)$$

The domain Ω is split in a collection of elements $K \in \mathcal{T}$.



$$(\partial_t \mathbf{w} - \mathbf{S}, w)_{\mathcal{T}} - (\mathbf{F}_c - \mathbf{F}_v, \nabla w)_{\mathcal{T}} + \sum_{K \in \mathcal{T}} \langle (\mathbf{F}_c - \mathbf{F}_v) \cdot \mathbf{n}, w \rangle_{\partial K} = 0, \quad \forall w \in W_h$$

We restrict w to a **finite dimensional** subset of $L_2(\Omega)$.

$$W_h = \{w \in L^2(\Omega) : w|_K \in \mathcal{P}^p(K), \forall K \in \mathcal{T}\} \subset L_2(\Omega)$$

$$(\partial_t \mathbf{w} - \mathbf{S}, w)_{\mathcal{T}} - (\mathbf{F}_c - \mathbf{F}_v, \nabla w)_{\mathcal{T}} + \sum_{K \in \mathcal{T}} \langle \mathcal{H}(\mathbf{u}, \nabla \mathbf{u}), w \rangle_{\partial K} = 0, \quad \forall w \in W_h$$

For stabilizing the method, the convective and diffusive fluxes on the interface are approximated

$$(\mathbf{F}_c - \mathbf{F}_v) \cdot \mathbf{n}_K \simeq \mathcal{H}$$

$$(\partial_t \mathbf{w}_h - \mathbf{S}_h, w)_{\mathcal{T}} - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla w)_{\mathcal{T}} + \sum_{K \in \mathcal{T}} \langle \mathcal{H}(\mathbf{u}, \nabla \mathbf{u}), w \rangle_{\partial K} = 0, \quad \forall w \in W_h$$

The solution is also approximated by

$$\mathbf{u} \simeq \mathbf{u}_h \in W_h$$

$$(\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla \varphi_{K,i})_K + \langle \mathcal{H}(\mathbf{u}_h, \nabla \mathbf{u}_h), \varphi_{K,i} \rangle_{\partial K} = 0, \\ \forall \varphi_{K,i} \in W_h$$

If the $\varphi_{K,i}$ is a local basis of W_h on K , $i \in \{1, 2, \dots, p\}$,

$$\mathbf{u} \simeq \mathbf{u}_h = \sum_{K \in \mathcal{T}} \sum_{i=1}^p \mathbf{u}_{K,i} \varphi_{K,i}$$

$$\begin{aligned} & (\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla \varphi_{K,i})_K + \langle \mathcal{H}(\mathbf{u}_h, \nabla \mathbf{u}_h), \varphi_{K,i} \rangle_{\partial K} = 0, \\ & \forall \varphi_{K,i} \in W_h \end{aligned}$$

- $\varphi_{K,i}$ are discontinuous across elements.
- If $\varphi_{K,i} = 1$ everywhere $\Rightarrow$ finite volume.

From weak form to HDG

$$\begin{aligned}(\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla \varphi_{K,i})_K + \langle \mathcal{H}(\mathbf{u}_h, \mathbf{q}_h), \varphi_{K,i} \rangle_{\partial K} &= 0, \\(\mathbf{q}_h, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_h, \nabla \boldsymbol{\tau}_{K,i})_K + \langle \mathbf{u}_h, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_{bc}} &= 0, \\ \forall \varphi_{K,i} \in W_h, \forall \boldsymbol{\tau}_{K,i} \in V_h\end{aligned}$$

With the solution gradient

$$\mathbf{q} = \nabla \mathbf{u} \simeq \mathbf{q}_h$$

with the subset

$$V_h = \left\{ \mathbf{v} \in L^2(\Omega) : \mathbf{v}|_K \in (\mathcal{P}^p(K))^D, \forall K \in \mathcal{T} \right\}$$

and local basis $\boldsymbol{\tau}_{K,i}$.

$$\begin{aligned}(\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla \varphi_{K,i})_K + \langle \mathcal{H}(\boldsymbol{\lambda}_h, \mathbf{u}_h, \mathbf{q}_h), \varphi_{K,i} \rangle_{\partial K} &= 0, \\(\mathbf{q}_h, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_h, \nabla \boldsymbol{\tau}_{K,i})_K + \langle \boldsymbol{\lambda}_h, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_{bc}} &= 0 \\ \forall \varphi_{K,i} \in W_h, \forall \boldsymbol{\tau}_{K,i} \in V_h\end{aligned}$$

$$\Gamma = \{e : e = K_i \cap K_j; \forall K_i, K_j \in \mathcal{T}, K_i \neq K_j\}.$$

$$M_h = \{\mu \in L^2(\Gamma) : \mu|_e \in \mathcal{P}^p(e), \forall e \in \Gamma\}$$

$$\boldsymbol{\lambda}_h = \sum_{e \in \Gamma} \sum_{i=1}^p \boldsymbol{\lambda}_{e,i} \mu_{e,i}$$

$$\begin{aligned}(\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla \varphi_{K,i})_K + \langle \mathcal{H}(\boldsymbol{\lambda}_h, \mathbf{u}_h, \mathbf{q}_h), \varphi_{K,i} \rangle_{\partial K} &= 0, \\(\mathbf{q}_h, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_h, \nabla \boldsymbol{\tau}_{K,i})_K + \langle \boldsymbol{\lambda}_h, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_{bc}} &= 0, \\ \langle [[\mathcal{H}]], \mu_{e,i} \rangle_e &= 0, \\ \forall \mu_{e,i} \in M_h, \forall \varphi_{K,i} \in W_h, \forall \boldsymbol{\tau}_{K,i} \in V_h\end{aligned}$$

Equation for $\boldsymbol{\lambda}$: conservativity of the normal numerical flux across the interfaces.

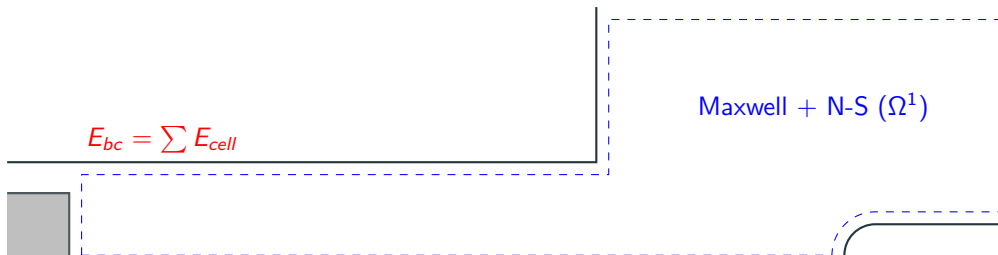
The jump operator is given by

$$[[\mathbf{u}]] = \mathbf{u}_+ \cdot \mathbf{n}_+ + \mathbf{u}_- \cdot \mathbf{n}_-$$

$$\begin{aligned}(\partial_t \mathbf{w}_h - \mathbf{S}_h, \varphi_{K,i})_K - (\mathbf{F}_{c,h} - \mathbf{F}_{v,h}, \nabla \varphi_{K,i})_K + \langle \mathcal{H}(\boldsymbol{\lambda}_h, \mathbf{u}_h, \mathbf{q}_h), \varphi_{K,i} \rangle_{\partial K} &= 0, \\(\mathbf{q}_h, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_h, \nabla \boldsymbol{\tau}_{K,i})_K + \langle \boldsymbol{\lambda}_h, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_{bc}} &= 0, \\ \langle [[\mathcal{H}]], \mu_{e,i} \rangle_e &= 0, \\ \forall \mu_{e,i} \in M_h, \forall \varphi_{K,i} \in W_h, \forall \boldsymbol{\tau}_{K,i} \in V_h\end{aligned}$$

ICP as a multi-domain problem

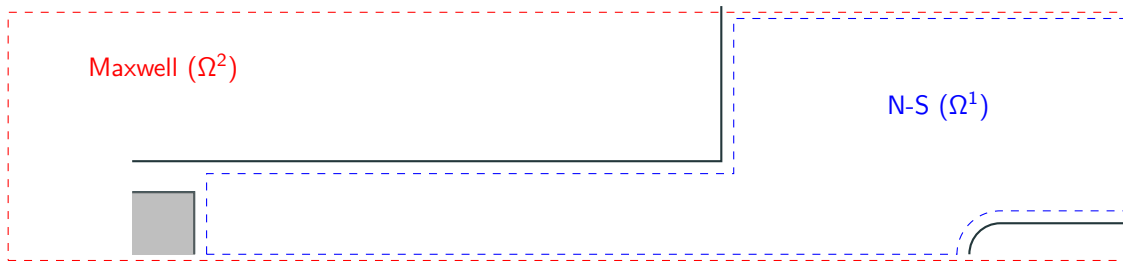
Single domain + integral boundary value



Compact but fills the jacobian matrix, making it incompatible with HDG.

ICP as a multi-domain problem

Two overlapping domain



Most widely used. Solve N-S while Maxwell frozen, and *vice versa*. Slow convergence.

ICP as a multi-domain problem

Multi-domain



Solves the system in a fully coupled manner, with two domains. The method we employ here.

Modification of the model problem

$$\partial_t \mathbf{w}^l(\mathbf{u}) + \nabla \cdot \left(\mathbf{F}_c^l(\mathbf{u}) - \mathbf{F}_v^l(\mathbf{u}, \nabla \mathbf{u}) \right) = \mathbf{S}^l(\mathbf{u}, \nabla \mathbf{u}), \quad \text{on } \Omega^l$$

$$\mathbf{u}^l = \mathbf{u}_{bc}^l, \quad \mathbf{x} \in \partial\Omega_d^l$$

$$\mathbf{F}_v^l \cdot \mathbf{n} = \mathbf{F}_{v,n,bc}^l, \quad \text{on } \partial\Omega_n^l$$

$$\mathcal{F}^{l,l'}(\mathbf{u}^l, \nabla \mathbf{u}^l, \mathbf{u}^{l'}, \nabla \mathbf{u}^{l'}) = 0, \quad \text{on } \gamma^{l,l'}, \quad l \neq l'$$

$$\mathbf{u}^l(t=0) = \mathbf{U}^l, \quad \text{on } \Omega^l$$

Modification of the model problem

$$\partial_t \mathbf{w}^l(\mathbf{u}) + \nabla \cdot \left(\mathbf{F}_c^l(\mathbf{u}) - \mathbf{F}_v^l(\mathbf{u}, \nabla \mathbf{u}) \right) = \mathbf{S}^l(\mathbf{u}, \nabla \mathbf{u}), \quad \text{on } \Omega^l$$

$$\mathbf{u}^l = \mathbf{u}_{bc}^l, \quad \mathbf{x} \in \partial\Omega_d^l$$

$$\mathbf{F}_v^l \cdot \mathbf{n} = \mathbf{F}_{v,n,bc}^l, \quad \text{on } \partial\Omega_n^l$$

$$\mathcal{F}^{l,l'}(\mathbf{u}^l, \nabla \mathbf{u}^l, \mathbf{u}^{l'}, \nabla \mathbf{u}^{l'}) = 0, \quad \text{on } \gamma^{l,l'}, \quad l \neq l'$$

$$\mathbf{u}^l(t=0) = \mathbf{U}^l, \quad \text{on } \Omega^l$$

Modification of the model problem

$$\begin{aligned}
 \partial_t \mathbf{w}^l(\mathbf{u}) + \nabla \cdot \left(\mathbf{F}_c^l(\mathbf{u}) - \mathbf{F}_v^l(\mathbf{u}, \nabla \mathbf{u}) \right) &= \mathbf{S}^l(\mathbf{u}, \nabla \mathbf{u}), & \text{on } \Omega^l \\
 \mathbf{u}^l &= \mathbf{u}_{bc}^l, & \mathbf{x} \in \partial\Omega_d^l \\
 \mathbf{F}_v^l \cdot \mathbf{n} &= \mathbf{F}_{v,n,bc}^l, & \text{on } \partial\Omega_n^l \\
 \mathcal{F}^{l,l'}(\mathbf{u}^l, \nabla \mathbf{u}^l, \mathbf{u}^{l'}, \nabla \mathbf{u}^{l'}) &= 0, & \text{on } \gamma^{l,l'}, l \neq l' \\
 \mathbf{u}^l(t=0) &= \mathbf{U}^l, & \text{on } \Omega^l
 \end{aligned}$$

For ICP,

$$\mathcal{F} = \begin{pmatrix} \nabla E_p^{Re,NS} \cdot \mathbf{n}^{NS} - \nabla E_p^{Re,MAX} \cdot \mathbf{n}^{MAX} \\ \nabla E_p^{Im,NS} \cdot \mathbf{n}^{NS} - \nabla E_p^{Im,MAX} \cdot \mathbf{n}^{MAX} \end{pmatrix}$$

Modification of the discrete formulation

$$\begin{aligned} & \left(\partial_t \mathbf{w}_h - \mathbf{S}_h^K, \varphi_{K,i} \right)_K - \left(\mathbf{F}_{h,c}^K - \mathbf{F}_{h,v}^K, \nabla \varphi_{K,i} \right)_K + \left\langle \mathcal{H}^K, \varphi_{K,i} \right\rangle_{\partial K} = 0 \\ & (\mathbf{q}_K, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_K, \nabla \boldsymbol{\tau}_{K,i})_K + \langle \boldsymbol{\lambda}_{\partial K}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_{bc}} = 0 \\ & \langle [[\mathcal{H}]], \mu_{e,i} \rangle_{e \setminus \bar{\Gamma}} + \langle \tilde{\mathcal{F}}^e, \mu_{e,i} \rangle_{e \cap \bar{\Gamma}} = 0 \end{aligned}$$

with $\bar{\Gamma}$ the set of interfaces between subdomains.

$$\mathcal{H} = \mathcal{H}_c - \mathcal{H}_v$$

$$\mathcal{H}_c = R\mathcal{H}_{c,n}(\mathbf{u}, \lambda) = R(\dot{m} + \dot{m}_p)\Psi_\lambda - R|\dot{m}|(\Psi_\lambda - \Psi_U) + \mathbf{P}$$

with

$$\boldsymbol{\Psi} = \left(1 \quad v^\perp \quad v^\parallel \quad v_\theta \quad e + \frac{1}{2}||\mathbf{v}||^2 + \frac{p}{\rho} \right)^T$$

$$\mathbf{P} = \left(0 \quad p_\lambda \quad 0 \quad 0 \right)^T$$

$$\mathcal{H} = \mathcal{H}_c - \mathcal{H}_v$$

$$\mathcal{H}_v(\mathbf{u}, \lambda, \mathbf{q}) = \mathbf{F}_v(\lambda, \mathbf{q}) \cdot \mathbf{n} + \gamma(h)(\lambda - \mathbf{u})$$

with

$$\gamma = C \max_{f \in K} \left(\frac{1}{2\mathcal{V}} \sum_{f \in K} \mathcal{A} \right)$$

C depends on the diffusive transport coefficients.

Solution strategy

The HDG system is solved at steady-state such that

$$\begin{aligned}
 & \left(\cancel{\partial_t \mathbf{w}_h} - \mathbf{S}_h^K, \varphi_{K,i} \right)_K - \left(\mathbf{F}_{h,c}^K - \mathbf{F}_{h,v}^K, \nabla \varphi_{K,i} \right)_K + \left\langle \mathcal{H}^K, \varphi_{K,i} \right\rangle_{\partial K} = 0 \\
 & (\mathbf{q}_K, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_K, \nabla \boldsymbol{\tau}_{K,i})_K + \langle \boldsymbol{\lambda}_{\partial K}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_{bc}} = 0 \\
 & \langle [[\mathcal{H}]], \mu_{e,i} \rangle_{e \setminus \bar{\Gamma}} + \langle \tilde{\mathcal{F}}^e, \mu_{e,i} \rangle_{e \cap \bar{\Gamma}} = 0
 \end{aligned}$$

If the system is written as $\mathcal{N} = 0$, then the Newton method solves

$$\mathcal{N}'(\mathbf{x}_k) \delta \mathbf{x}_k = -\mathcal{N}(\mathbf{x}_k)$$

with $\delta \mathbf{x}_k = \mathbf{x}_{k+1} - \mathbf{x}_k$.

Solution strategy

The HDG system is solved at steady-state such that

$$\begin{aligned} \left(\frac{\delta \mathbf{w}_h^K}{\tau_{K,k}} - \mathbf{S}_h^K, \varphi_{K,i} \right)_K - \left(\mathbf{F}_{h,c}^K - \mathbf{F}_{h,v}^K, \nabla \varphi_{K,i} \right)_K + \left\langle \mathcal{H}^K, \varphi_{K,i} \right\rangle_{\partial K} &= 0 \\ (\mathbf{q}_K, \boldsymbol{\tau}_{K,i})_K - (\mathbf{u}_K, \nabla \boldsymbol{\tau}_{K',i})_K + \langle \boldsymbol{\lambda}_{\partial K}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_0} + \langle \mathbf{u}_{bc}, \boldsymbol{\tau}_{K,i} \cdot \mathbf{n} \rangle_{\partial K_{bc}} &= 0 \\ \langle [[\mathcal{H}]], \mu_{e,i} \rangle_{e \setminus \bar{\Gamma}} + \langle \tilde{\mathcal{F}}^e, \mu_{e,i} \rangle_{e \cap \bar{\Gamma}} &= 0 \end{aligned}$$

If the system is written as $\tilde{\mathcal{N}} = 0$, then the Newton method solves

$$\tilde{\mathcal{N}}'(\mathbf{x}_k) \delta \mathbf{x} = -\tilde{\mathcal{N}}(\mathbf{x}_k)$$

with $\delta \mathbf{x} = \mathbf{x}_{k+1} - \mathbf{x}_k$.

In order to stabilize the solver, a fictive temporal term is added.

Damping term

$$\frac{\delta \mathbf{w}_h}{\tau_K} = \frac{\mathbf{w}(\mathbf{u}_{h,k+1}) - \mathbf{w}(\mathbf{u}_{h,k})}{\tau_{K,k}}$$

with

$$\tau_{K,k} = CFL_k \frac{h_K}{v_K} = CFL_k \frac{V_K}{\int_{\partial K} \lambda_{\max} dS}$$

and the following CFL evolution law

$$CFL_k = \min \left(CFL_0 \left(\frac{||\tilde{\mathcal{N}}||_2^0}{||\tilde{\mathcal{N}}||_2^k} \right)^\alpha, CFL_{\max} \right)$$

Damping term

$$\frac{\delta \mathbf{w}_h}{\tau_K} = \frac{\mathbf{w}(\mathbf{u}_{h,k+1}) - \mathbf{w}(\mathbf{u}_{h,k})}{\tau_{K,k}}$$

with

$$\tau_{K,k} = CFL_k \frac{h_K}{v_K} = CFL_k \frac{V_K}{\int_{\partial K} \lambda_{\max} dS}$$

and the following CFL evolution law

$$CFL_k = \min \left(CFL_0 \left(\frac{||\tilde{\mathcal{N}}||_2^0}{||\tilde{\mathcal{N}}||_2^k} \right)^\alpha, CFL_{\max} \right)$$

We only use convective eigenvalues here. The diffusive could be use to enhance the convergence.

Matrix form of the system

$$\tilde{\mathcal{N}}'(\mathbf{x}_k)\delta\mathbf{x}_k = -\tilde{\mathcal{N}}(\mathbf{x}_k)$$

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$$\underbrace{\begin{pmatrix} A & B & R \\ C & D & S \\ L & M & N \end{pmatrix}}_{\tilde{\mathcal{N}}'} \overbrace{\begin{pmatrix} \delta Q \\ \delta U \\ \delta \Lambda \end{pmatrix}}^{\delta \mathbf{x}} = \underbrace{\begin{pmatrix} F \\ G \\ H \end{pmatrix}}_{-\tilde{\mathcal{N}}},$$

Matrix form of the system

$$\tilde{\mathcal{N}}'(\mathbf{x}_k)\delta\mathbf{x}_k = -\tilde{\mathcal{N}}(\mathbf{x}_k)$$

The system can be split into two parts

$$\underbrace{\begin{pmatrix} A & B \\ C & D \end{pmatrix}}_{\text{Block diagonal}} \begin{pmatrix} \delta Q \\ \delta U \end{pmatrix} = \begin{pmatrix} F \\ G \end{pmatrix} - \begin{pmatrix} R \\ S \end{pmatrix} \delta\Lambda$$

and

$$\begin{pmatrix} L & M \end{pmatrix} \begin{pmatrix} \delta Q \\ \delta U \end{pmatrix} + N\delta\Lambda = H$$

Matrix form of the system

$$\tilde{\mathcal{N}}'(\mathbf{x}_k)\delta\mathbf{x}_k = -\tilde{\mathcal{N}}(\mathbf{x}_k)$$

By eliminating δU and δQ , one gets the **global** system

$$\left(N - \begin{pmatrix} L & M \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} R \\ S \end{pmatrix} \right) \delta\Lambda = H - \begin{pmatrix} L & M \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} F \\ G \end{pmatrix}$$

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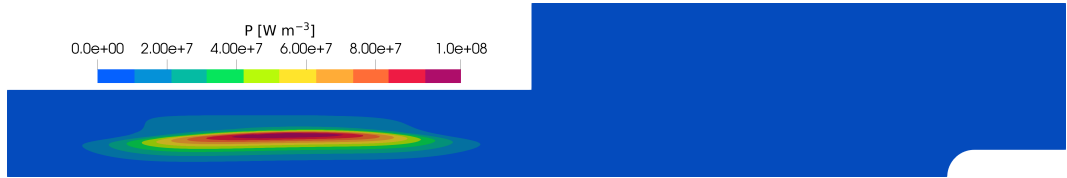
$$\left(N - \begin{pmatrix} L & M \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} R \\ S \end{pmatrix} \right) \delta\Lambda = H - \begin{pmatrix} L & M \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} F \\ G \end{pmatrix}$$

System solved for λ , then $\mathbf{u}$ and $\mathbf{q}$ are retrieved by small local matrices inversion.

$$\begin{pmatrix} \delta Q \\ \delta U \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} F \\ G \end{pmatrix} - \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} \begin{pmatrix} R \\ S \end{pmatrix} \delta\Lambda$$

The problem of power quenching

If the power dissipated is not maintained, the torch quenches!



$$P_{target} = \int_{NS} P_J dV$$

The problem of power quenching

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$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v} \mathbf{v}) + \nabla p - \nabla \cdot \boldsymbol{\tau} = \mathbf{F}^L$$

$$\partial_t \left(\rho e + \frac{1}{2} \rho \|\mathbf{v}\|^2 \right) + \nabla \cdot \left(\rho e \mathbf{v} + \frac{1}{2} \rho \|\mathbf{v}\|^2 \mathbf{v} + p \mathbf{v} - \boldsymbol{\tau} \cdot \mathbf{v} - \mathbf{q} \right) = P_J$$

$$\Delta E_P = \frac{E_P}{r^2} + i2\pi f \mu_0 \sigma_e (E_C + E_P)$$

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$$\partial_t \left(\rho e + \frac{1}{2} \rho \|\mathbf{v}\|^2 \right) + \nabla \cdot \left(\rho e \mathbf{v} + \frac{1}{2} \rho \|\mathbf{v}\|^2 \mathbf{v} + p \mathbf{v} - \boldsymbol{\tau} \cdot \mathbf{v} - \mathbf{q} \right) = \gamma P_J'$$

$$\Delta E_P' = \frac{E_P'}{r^2} + i 2 \pi f \mu_0 \sigma_e (E_C' + E_P')$$

with

$$E_P = \sqrt{\gamma} E_P' \quad E_C = \sqrt{\gamma} E_C'$$

γ is computed at every Newton iteration by equalizing

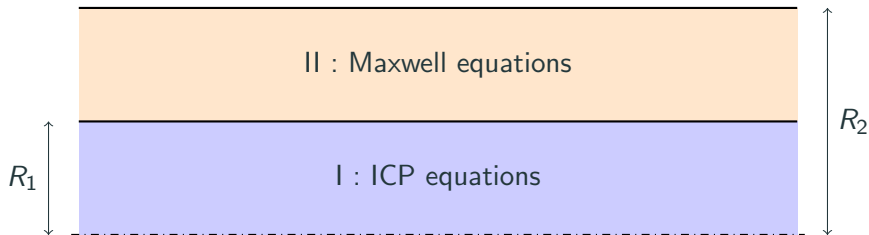
$$\gamma = \frac{P_{target}}{\int_{NS} P'_J dV}$$

with P_{target} the desired power dissipated in the facility.

The power was found to remain constant and γ was converging towards a value using this method.

Results

Manufactured solution



$$p = \Delta p_0 f(z, r) + p_0 \quad T = T_{min} + (T_{max} - T_{min}) f(z, r)$$

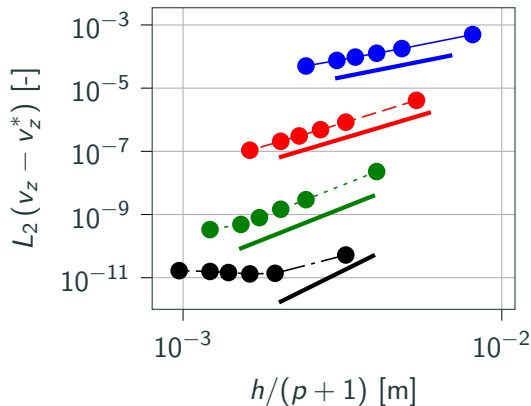
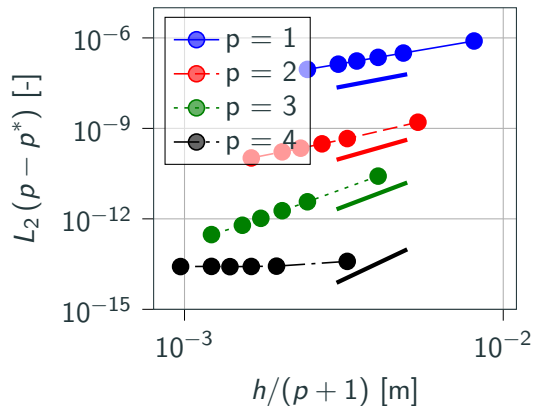
$$v_z = u_0 f(z, r) \quad E_p = E_0 f(z, r)(1 - i)$$

$$v_r = u_0 f(z, r) \quad v_\theta = u_0 f(z, r)$$

with

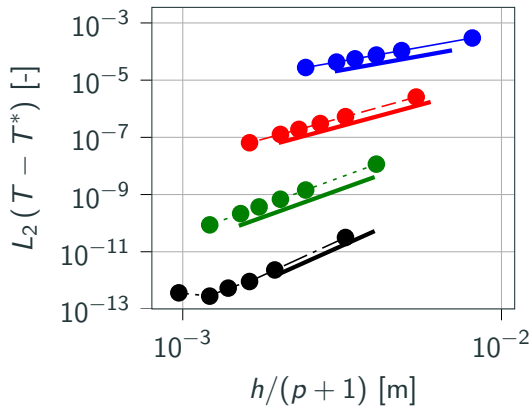
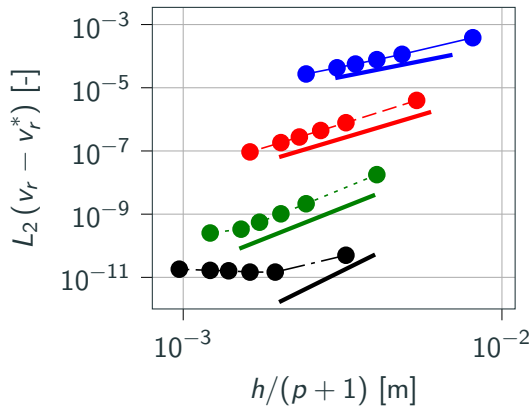
$$f(z, r) = \left(\frac{rz}{RL} \right)^2 \exp \left(-\frac{z}{L} - \frac{r}{R_1} \right)$$

Convergence study : in the torch



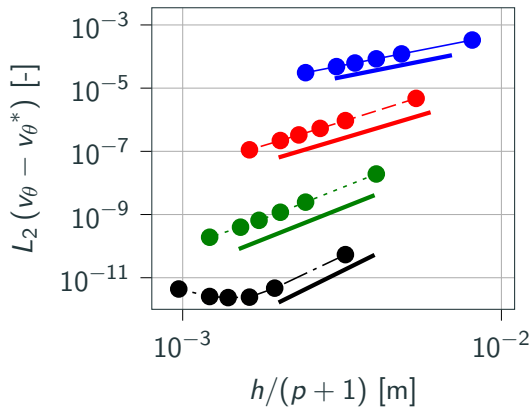
Convergence order of $p + 1$ is retrieved!

Convergence study : in the torch



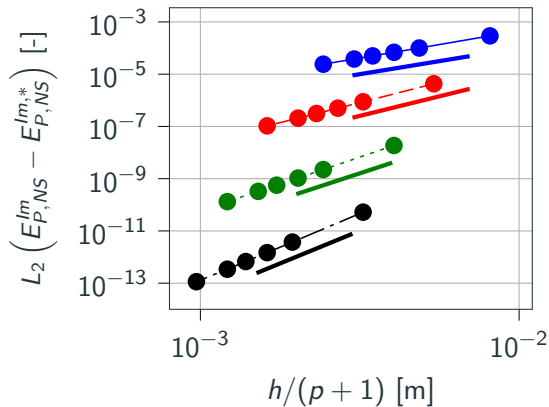
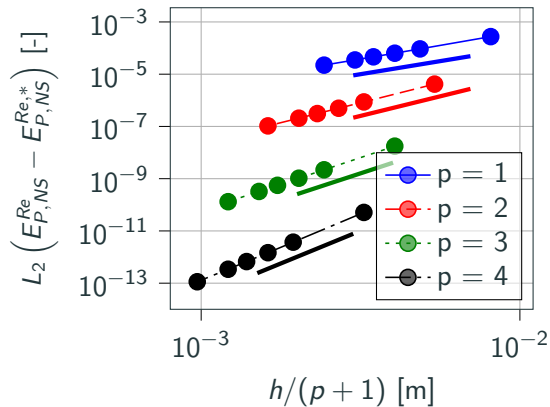
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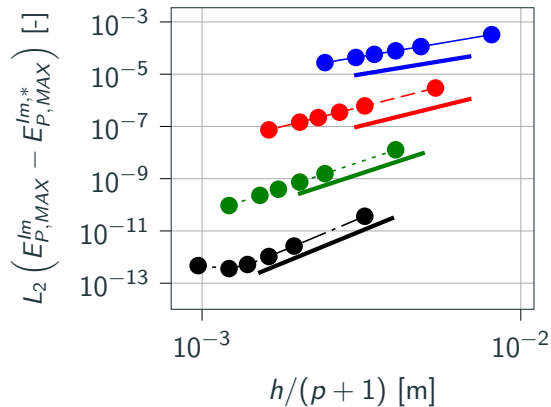
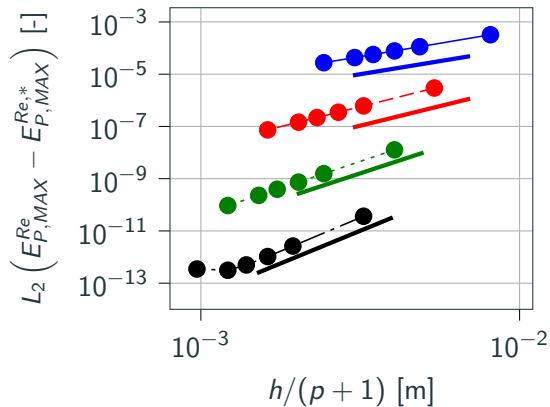
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Convergence study : outside the torch



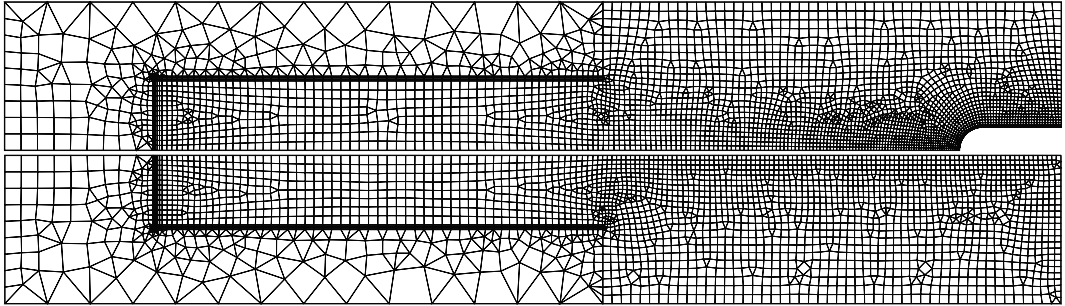
Convergence order of $p+1$ is retrieved!

Convergence study : outside the torch



Convergence order of $p + 1$ is retrieved!

Mesh and order dependence study



This mesh at $p = 4$ with 4013 elements for the probe case and 3055 for the freestream case is sufficient for capturing the solution accurately.

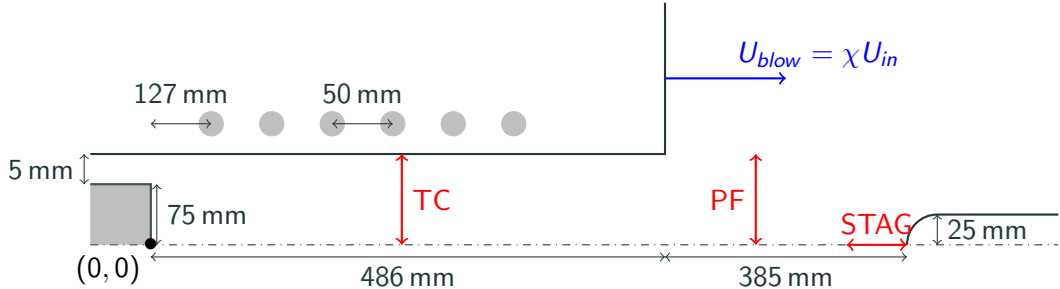
The mesh size close to the probe is $10\text{ }\mu\text{m}$ thick, while it is of $1\text{ }\mu\text{m}$ for FV!

The question of the coflow

A coflow is introduced in the chamber for obtaining a steady state, does it have an important impact?

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A coflow is introduced in the chamber for obtaining a steady state, does it have an important impact?

- $P = 100 \text{ kW}$
- $p_0 = 10\,000 \text{ Pa}$
- $f = 0.37 \text{ MHz}$
- $Q = 16 \text{ g s}^{-1}$
- $S = 20^\circ$
- air mixture with 11 species

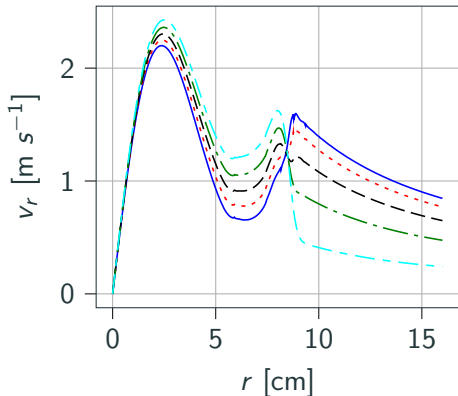
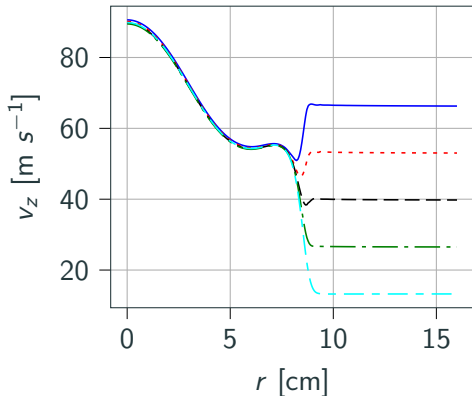
The coflow χ is defined as

$$\chi = \frac{U_{blow}}{U_{in}}$$

and tested from 0.2 to 1.0.

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The question of the coflow

A coflow is introduced in the chamber for obtaining a steady state, does it have an important impact? Short answer is no, BUT...

... we were not able to obtain any steady state without coflow.

A question to ask is to see if steady state is possible in ICP?

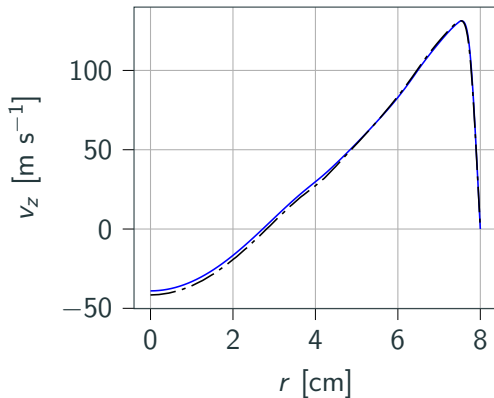
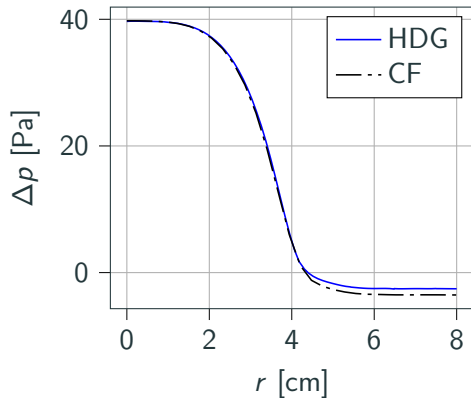
Left for future work...

We compared the results of the COOLFluid solver and the HDG code.

- $p_0 = 5000 \text{ Pa}$
- $Q = 16 \text{ g s}^{-1}$
- $T_{wall} = T_{inlet} = 350 \text{ K}$
- $P = 100 \text{ kW}$
- $f = 0.37 \text{ MHz}$
- air mixture with 11 species.
- No swirl angle.
- The coflow applied for HDG is $\chi_{HDG} = 0.2$.

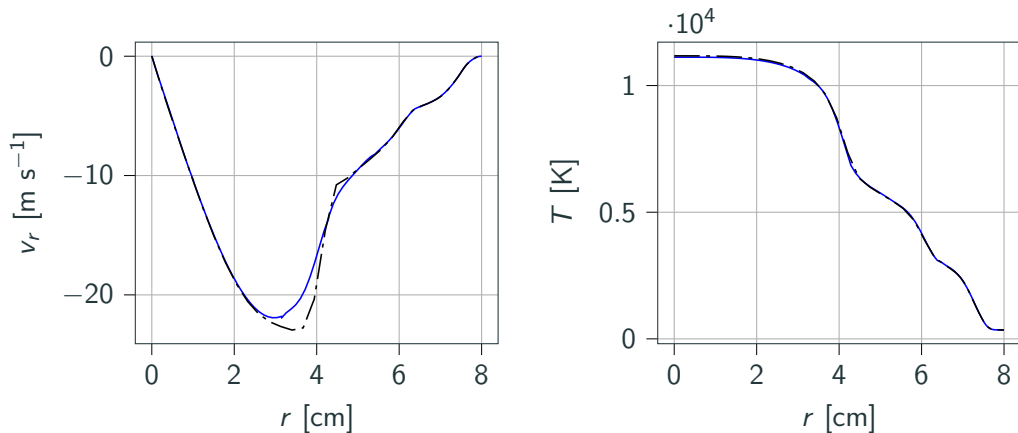
Comparison with the COOLFluid solver

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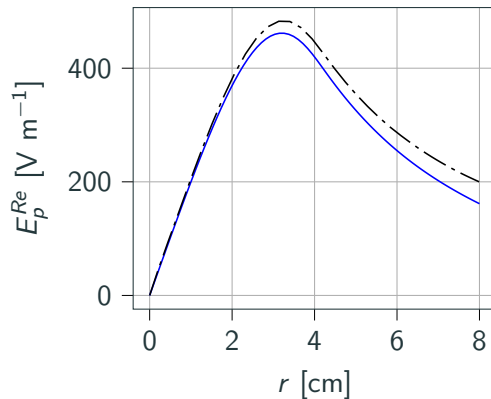
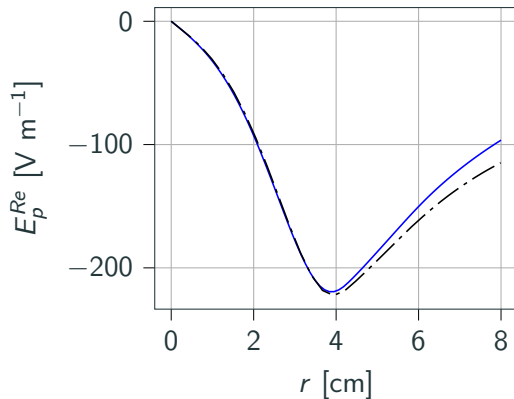
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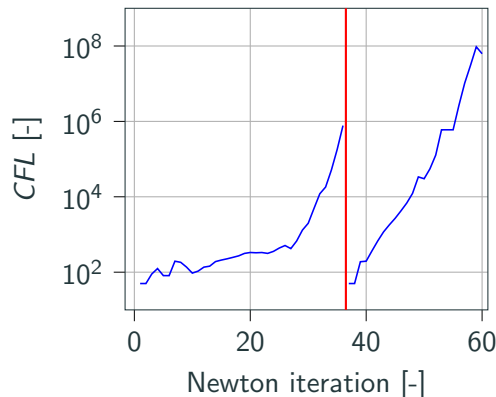
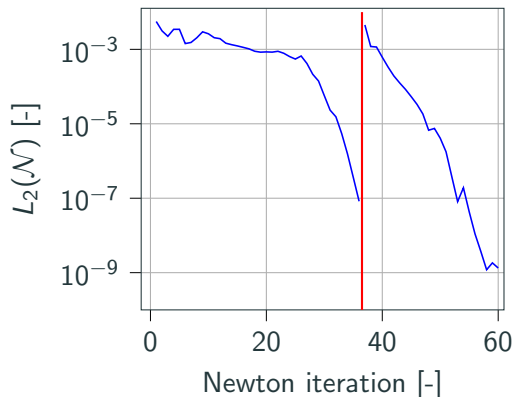
We compared the results of the COOLFluid solver and the HDG code.

Very similar results except for the electric field in the torch, might be due to the way electric field boundary conditions are imposed.

Convergence history

We use **GMRES** solver with at most 50 vectors and a **ILU(4)** preconditioner.

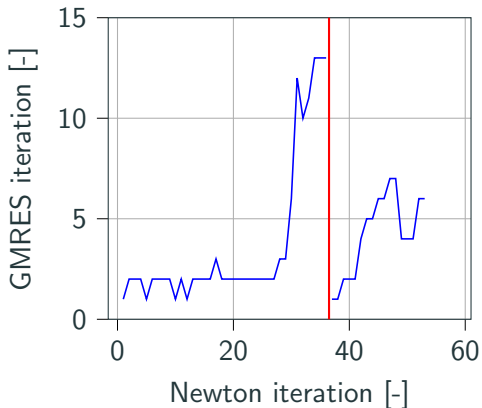
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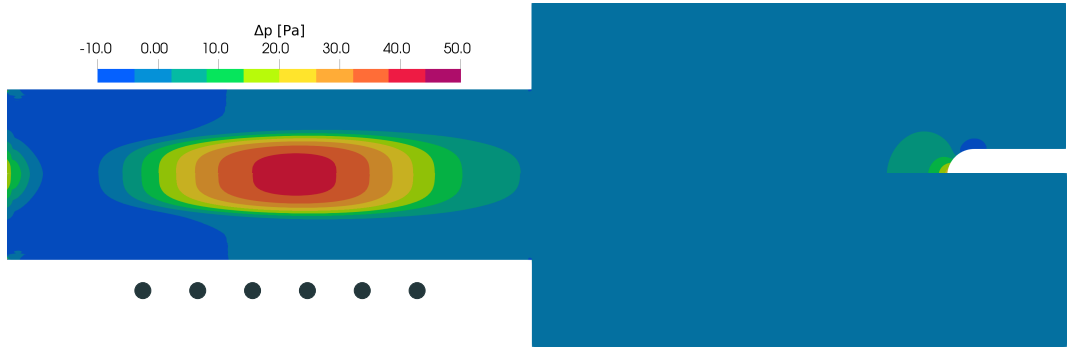
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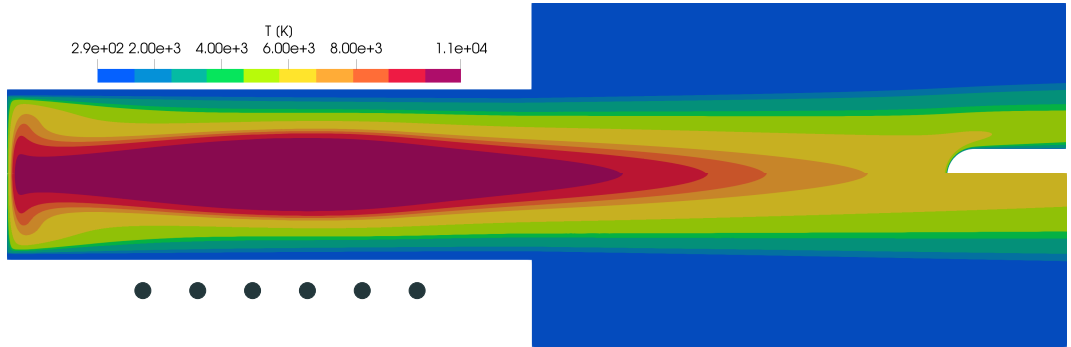
We have a fast and robust solver for ICP, able to produce meaningful results in 20 minutes of time on a single laptop using 8 OMP threads!

Flow fields in ICP: gauge pressure



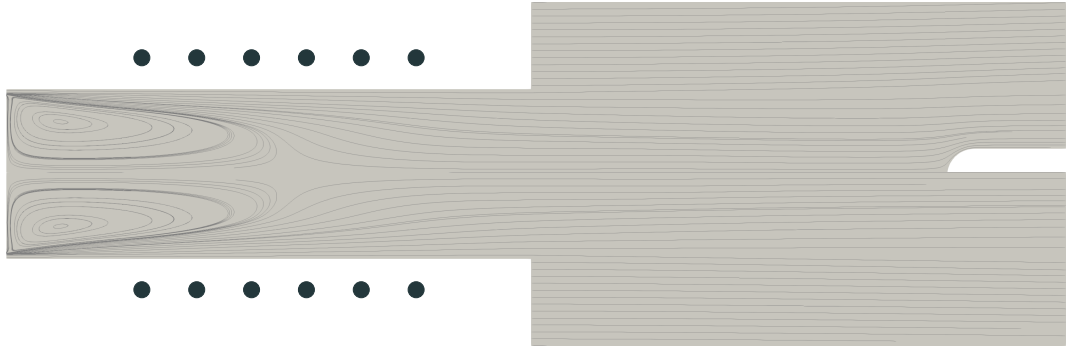
The pressure is almost constant everywhere.

Flow fields in ICP: temperature



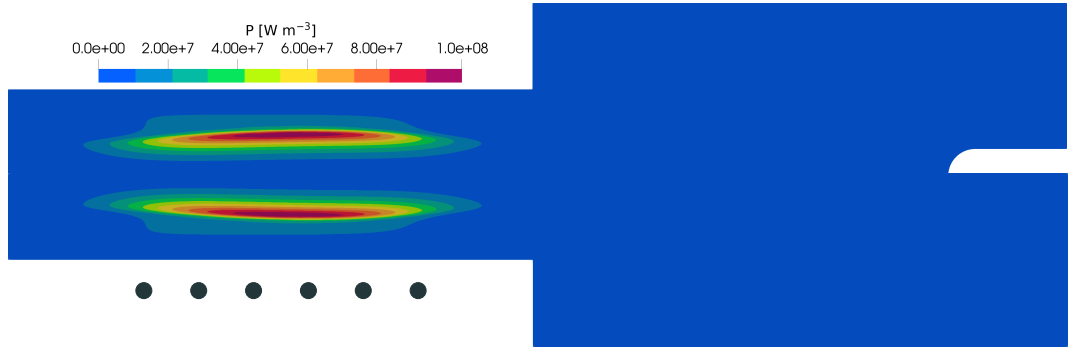
Large temperature gradients.

Flow fields in ICP: streamlines



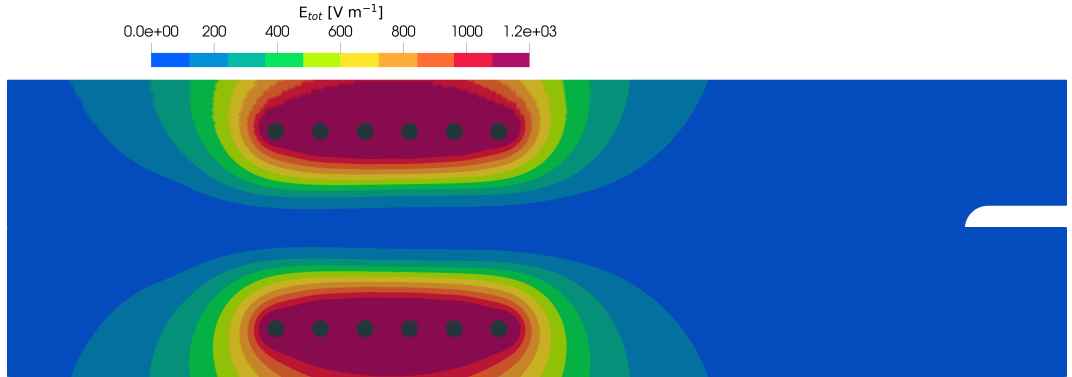
$$\nabla \cdot \mathbf{v} \simeq 0$$

Flow fields in ICP: volume power

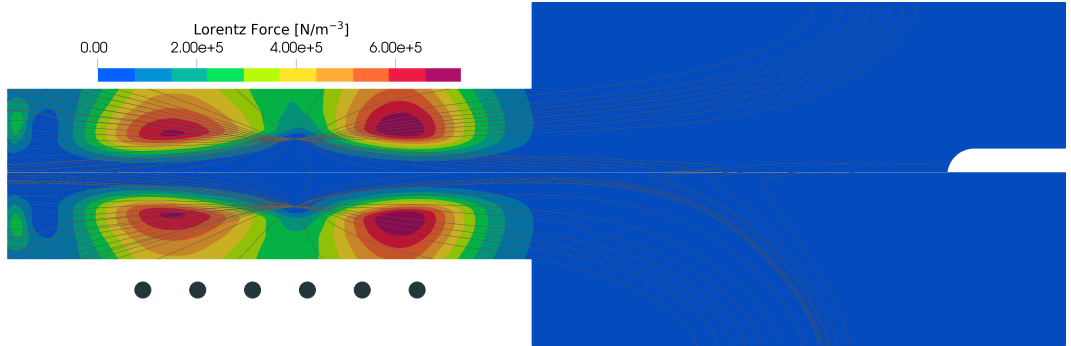


Volume power peak off-axis, where the coupling is the strongest.

Flow fields in ICP: Total electric field



Flow fields in ICP: Lorentz force



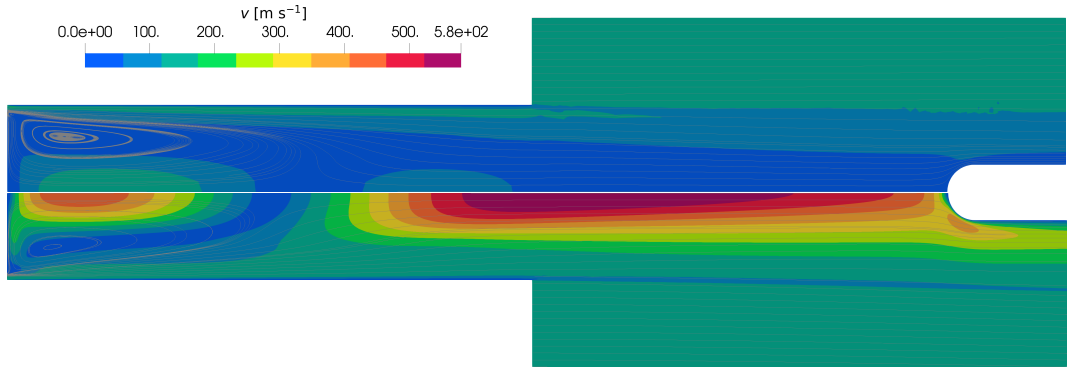
Parametric study

We performed a parametric study of ICP for the parameters used by the experimenters.

- Q from 8 g s^{-1} to 16 g s^{-1} .
- p_0 from 5000 Pa to 20 000 Pa.
- P_{tot} from 50 kW to 200 kW.
- f from 10 kHz, to 1 MHz.
- S from 0° to 20° .

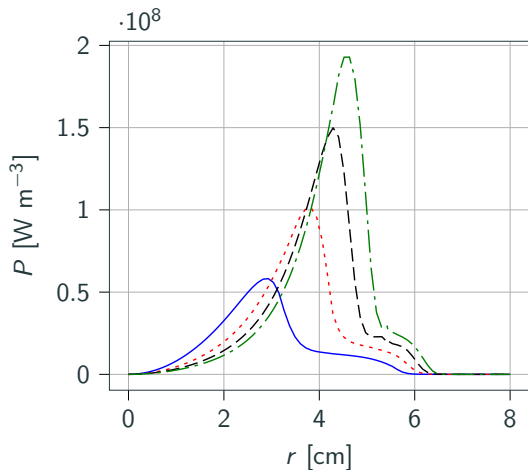
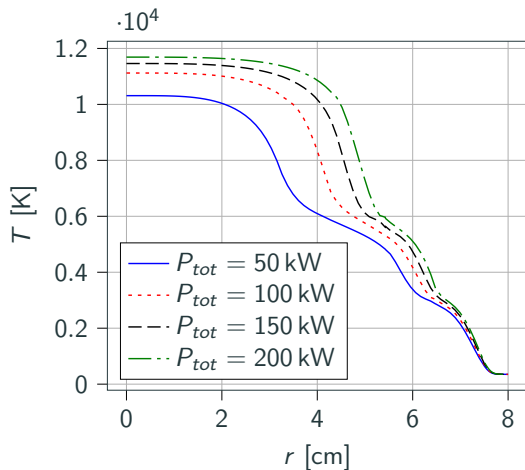
The complete discussion is in the thesis.

Power dissipated in the facility: impact on mass flow rate

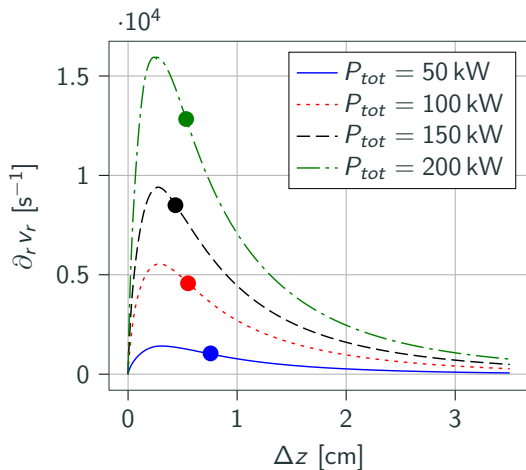
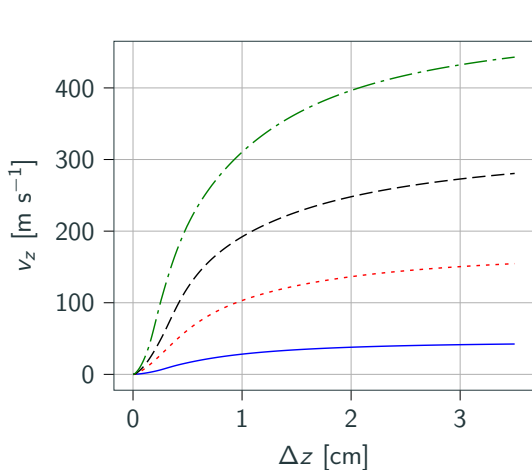


$$P_{tot} = 50 \text{ kW (top) and } P_{tot} = 200 \text{ kW (bottom)}$$

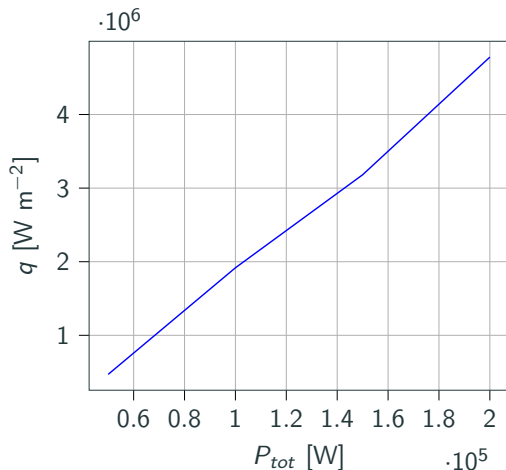
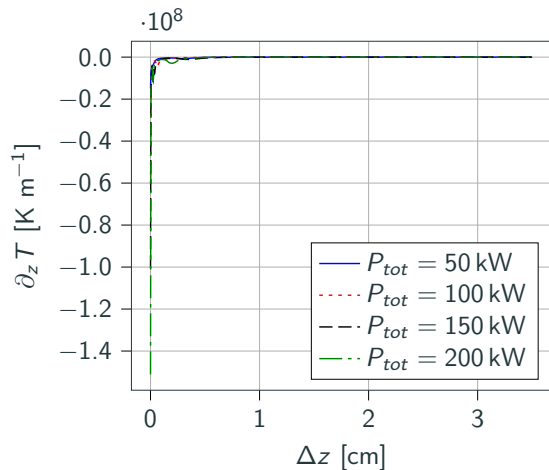
Power dissipated in the facility: impact in the torch



Power dissipated in the facility: impact along the stagnation line



Power dissipated in the facility: impact along the stagnation line



Conclusions and future work

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However, much easier to use and give meaningful results fast.

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Q3 Can the new solver be easily adapted to the new experiments performed in ICP facilities?

We believe so, but there is still much work to do...

Going 2D, 3D, and complex chemistry and radiation for unsteady simulations will need much more computational power.

For supersonic, shock capturing and redesign of the numerical flux.

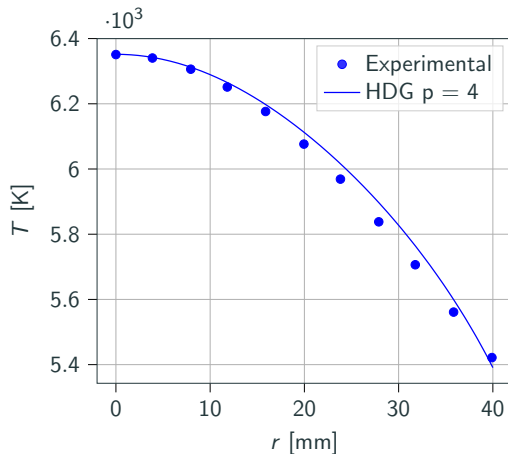
Future work

Main effort is to put our developments in the ARGO DG solver, which is fully paralleled and equipped shock capturing technologies. (Landry Riou)

We see the following roadmap:

- Going 3D.
- Going unsteady (also relieving the average over the induction current period).
- Development of chemistry (non equilibrium).
- Development of radiation.
- For all of the above, comparison with experiment when possible.

Comparison with experiment



- $p_0 = 200 \text{ mbar}$
- $Q = 16 \text{ g s}^{-1}$
- $P_{exp.} = 200 \text{ kW}$
- $P_{num.} = 78 \text{ kW}$
- Freestream, no swirl, air.

Comparison of the experimental $T(r)$ at 385 mm from the torch exit, from $r = 0$ to 40 mm.